ROB 5014 Math and Simulation for Robotics
Lecture: Forward Kinematics

Hyungpil Moon, Professor

SUNG KYUN KWAN UNIVERSITY (SKKU)

Department of Mechanical Engineering
Department of Intelligent Robotics

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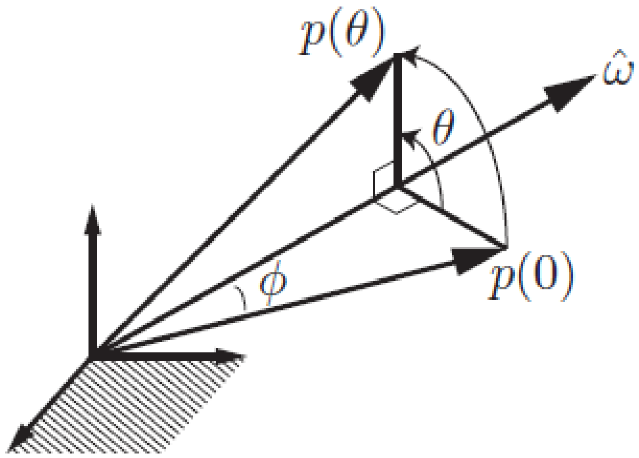
Linear Differential Equation

- Equation of motion
  - $\dot{x}(t) = a\,x(t)$
- Solution
  - $x(t) = e^{at}x_0$
- Computation of the exponential
  - $e^{at} = 1 + at + \frac{(at)^2}{2!} + \frac{(at)^3}{3!} + \cdots$
- If multiple variables
  - $x(t) = e^{At}x_0$
  - $e^{At} = I + At + \frac{(At)^2}{2!} + \frac{(At)^3}{3!} + \cdots$

Exponential Coordinates of Rotations

- The rotation is achieved by that $p(0)$ rotates at a constant rate of 1 rad/s from $t = 0$ to $t = \theta$ along $\hat{\omega}$.
- Equation of motion
  - $\dot{p} = \hat{\omega} \times p$
  - $\dot{p} = [\hat{\omega}]p$
- Solution
  - $p(t) = e^{[\hat{\omega}]t}p(0)$ or more properly, $p(\theta) = e^{[\hat{\omega}]\theta}p(0)$
- Note that
  - $e^{[\hat{\omega}]\theta} = I + [\hat{\omega}]\theta + \frac{([\hat{\omega}]\theta)^2}{2!} + \frac{([\hat{\omega}]\theta)^3}{3!} + \cdots$
  - As $[\hat{\omega}]^3 = -[\hat{\omega}]$,
  - $e^{[\hat{\omega}]\theta} = I + \left(\theta - \frac{\theta^3}{3!} + \frac{\theta^5}{5!} - \cdots\right)[\hat{\omega}] + \left(\frac{\theta^2}{2!} - \frac{\theta^4}{4!} + \frac{\theta^6}{6!} - \cdots\right)[\hat{\omega}]^2$
  - Because $\sin\theta = \theta - \frac{\theta^3}{3!} + \frac{\theta^5}{5!} - \cdots$, $\cos\theta = 1 - \frac{\theta^2}{2!} + \frac{\theta^4}{4!} - \frac{\theta^6}{6!} - \cdots$
- Rodrigues' formula
  - $e^{[\hat{\omega}]\theta} = I + \sin\theta\,[\hat{\omega}] + (1 - \cos\theta)[\hat{\omega}]^2$



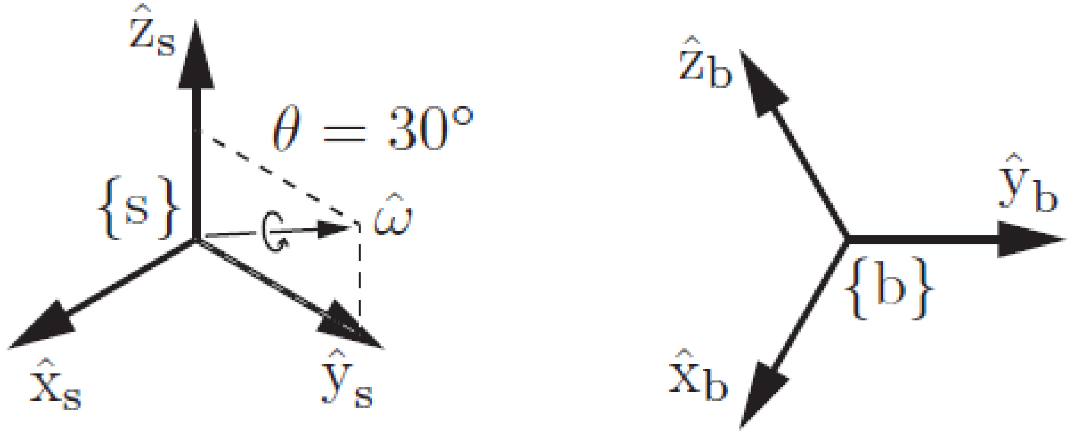
Exponential Coordinates of Rotations: Example

- Rodrigues' formula
  - $e^{[\hat{\omega}]\theta} = I + \sin\theta\,[\hat{\omega}] + (1 - \cos\theta)[\hat{\omega}]^2$
- The rotation matrix representation of {b}

$$\begin{aligned} R &= e^{[\hat{\omega}_1]\theta_1} \\ &= I + \sin\theta_1 [\hat{\omega}_1] + (1-\cos\theta_1)[\hat{\omega}_1]^2 \\ &= I + 0.5 \begin{bmatrix} 0 & -0.5 & 0.866 \\ 0.5 & 0 & 0 \\ -0.866 & 0 & 0 \end{bmatrix} + 0.134 \begin{bmatrix} 0 & -0.5 & 0.866 \\ 0.5 & 0 & 0 \\ -0.866 & 0 & 0 \end{bmatrix}^2 \\ &= \begin{bmatrix} 0.866 & -0.250 & 0.433 \\ 0.250 & 0.967 & 0.058 \\ -0.433 & 0.058 & 0.899 \end{bmatrix}. \end{aligned}$$

- Exponential coordinates
  - $\hat{\omega}_1\theta_1 = (0,\ 0.453,\ 0.262)$

The frame {b} is obtained by a rotation from {s} by $\theta_1 = 30° = 0.524\,rad$ about $\hat{\omega}_1 = (0, 0.866, 0.5)$



Exponential Coordinates of Rotations: Consecutive

- Case 1
  - $R = e^{[\hat{\omega}_1]\theta_1}$, then rotated by θ_2 about a fixed−frame axis $\hat{\omega}_2 \neq \hat{\omega}_1$
  - $R' = e^{[\hat{\omega}_2]\theta_2} R$:
- Case 2
  - $R = e^{[\hat{\omega}_1]\theta_1}$, then rotated by θ_2 about an axis expressed as $\hat{\omega}_2$ in body frame
  - $R'' = R\, e^{[\hat{\omega}_2]\theta_2} \neq R'$

Matrix Logarithm of Rotations

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- For the given rotational axis and angle, we can compute the rotation matrix
  - This is exponential coordinate of rotation
- For the given rotation matrix, we can compute the rotational axis and angle
  - This is logarithm of rotations

$$
\begin{aligned}
\exp : \quad & [\hat{\omega}]\theta \in so(3) \quad \rightarrow \quad R \in SO(3), \\
\log : \quad & R \in SO(3) \quad \rightarrow \quad [\hat{\omega}]\theta \in so(3).
\end{aligned}
$$

Algorithm for Matrix Logarithm of Rotations

- Problem: Given $R \in SO(3)$, find a $\theta \in [0, \pi]$ and a unit rotation axis $\hat{\omega} \in \Re^3, \left|\left|\hat{\omega}\right|\right| = 1$ such that $e^{[\hat{\omega}]\theta} = R$.
  - The vector $\hat{\omega}\theta \in \Re^3$ comprises the exponential coordinates of R
  - The skew-symmetric matrix $[\hat{\omega}]\theta \in so(3)$ is the matrix logarithm of R

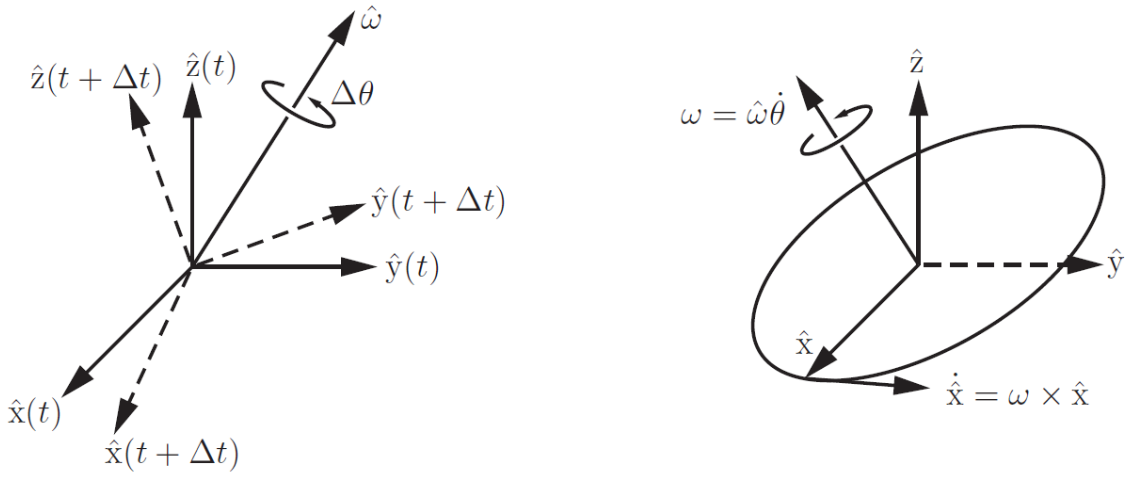
1) If $R = I$, then $\theta = 0$ and $\hat{\omega}$ is undefined.
2) If $tr(R) = -1$, then $\theta = \pi$ and set $\hat{\omega}$ among the following

$$\hat{\omega} = \frac{1}{\sqrt{2(1+r_{33})}}\begin{bmatrix} r_{13} \\ r_{23} \\ 1 + r_{33} \end{bmatrix} \text{ or } \hat{\omega} = \frac{1}{\sqrt{2(1+r_{22})}}\begin{bmatrix} r_{12} \\ 1 + r_{22} \\ r_{32} \end{bmatrix} \text{ or } \hat{\omega} = \frac{1}{\sqrt{2(1+r_{11})}}\begin{bmatrix} 1 + r_{11} \\ r_{21} \\ r_{31} \end{bmatrix}$$

3) Otherwise, $\theta = \cos^{-1}\left(\frac{1}{2}(tr(R) - 1)\right) \in [0, \pi)$ and $[\hat{\omega}] = \frac{1}{2\sin\theta}(R - R^T)$

Angular Velocities

- $\{\hat{x}, \hat{y}, \hat{z}\}$ is attached to a rotating body
- Angular velocity $w = \hat{w}\dot{\theta}$
- Velocity of the body
  - $\dot{\hat{x}} = w \times \hat{x}$
  - $\dot{y} = w \times \hat{y}$
  - $\dot{\hat{z}} = w \times \hat{z}$
- Reference frames
  - Fixed frame {s}
  - Body frame {b} which is attached to the body
- R(t) the rotation matrix describing the orientation of {b} w.r.t. {s}
- Then, $\dot{r}_i = w_s \times r_i$ (r_1: $\hat{x}$ in {s} coordinates, w_s: ang. Vel. expressed in {s})
- Thus, $\dot{R}(t) = w_s \times R = [w_s]R \rightarrow [w_s] = \dot{R}R^T$
- Note that $w_s = Rw_b$, $R[w]R^T = [Rw]$
- thus $w_b = R^T w_s$, $[w_b] = [R^T w_s] = R^T[w_s]R = R^T\dot{R}$



Twists

- Fixed/spatial coordinates {s}
- Moving/body coordinates {b}
- The configuration of {b} seen from {s} $T = T_{sb}(t) = \begin{bmatrix} R(t) & p(t) \\ 0 & 1 \end{bmatrix}$
- For rotation, we had $[w_s] = \dot{R}R^T, [w_b] = R^T\dot{R}$
- Likewise

$$\begin{aligned} \dot{T}T^{-1} &= \begin{bmatrix} \dot{R} & \dot{p} \\ 0 & 0 \end{bmatrix} \begin{bmatrix} R^{\mathrm{T}} & -R^{\mathrm{T}}p \\ 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} \dot{R}R^{\mathrm{T}} & \dot{p} - \dot{R}R^{\mathrm{T}}p \\ 0 & 0 \end{bmatrix} \\ &= \begin{bmatrix} [\omega_s] & v_s \\ 0 & 0 \end{bmatrix}. \end{aligned}$$

$$\begin{aligned} T^{-1}\dot{T} &= \begin{bmatrix} R^{\mathrm{T}} & -R^{\mathrm{T}}p \\ 0 & 1 \end{bmatrix} \begin{bmatrix} \dot{R} & \dot{p} \\ 0 & 0 \end{bmatrix} \\ &= \begin{bmatrix} R^{\mathrm{T}}\dot{R} & R^{\mathrm{T}}\dot{p} \\ 0 & 0 \end{bmatrix} \\ &= \begin{bmatrix} [\omega_b] & v_b \\ 0 & 0 \end{bmatrix}. \end{aligned}$$

Physical Meaning

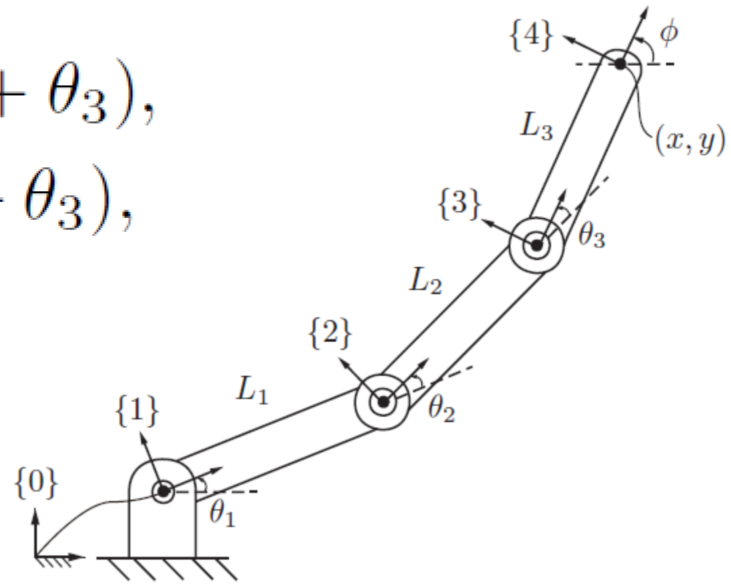
- Body twist, the spatial velocity in the body frame
  - $V_b = \begin{bmatrix} \omega_b \\ v_b \end{bmatrix} \in \Re^6$, $[V_b] = T^{-1}\dot{T} = \begin{bmatrix} [\omega_b] & v_b \\ 0 & 0 \end{bmatrix} \in se(3)$
- In rotation, $[w_s] = \dot{R}R^T$ is the angular velocity in {s}
- Is $v_s = \dot{p} - \dot{R}R^T p$ is the linear velocity of the origin of the body frame in {s} ?
  - The linear velocity of the body frame origin in {s} is simply $\dot{p}$
- If we use $v_s = \dot{p} - w_s \times p = \dot{p} + w_s \times (-p)$, we can read its physical meaning:
  - Instantaneous velocity of the point on the body currently at the origin of {s} expressed in {s}
- Spatial twist, the spatial velocity in {s}
  - $V_s = \begin{bmatrix} \omega_s \\ v_s \end{bmatrix} \in \Re^6, [V_s] = \dot{T}T^{-1} \in se(3)$
- w_b : ang vel expressed in {b}, w_s : ang vel expressed in {s}
- v_b : lin vel of a point at the origin of {b} expressed in {b}
- v_s : lin vel of a point at the origin of {s} expressed in {s}

$$
\begin{aligned}
\dot{T}T^{-1} &= \begin{bmatrix} \dot{R} & \dot{p} \\ 0 & 0 \end{bmatrix} \begin{bmatrix} R^{\mathrm{T}} & -R^{\mathrm{T}}p \\ 0 & 1 \end{bmatrix} \\
&= \begin{bmatrix} \dot{R}R^{\mathrm{T}} & \dot{p} - \dot{R}R^{\mathrm{T}}p \\ 0 & 0 \end{bmatrix} \\
&= \begin{bmatrix} [\omega_s] & v_s \\ 0 & 0 \end{bmatrix}.
\end{aligned}
$$

Forward Kinematics: Homogeneous Transformation

- Position

$$
\begin{aligned}
x &= L_1 \cos\theta_1 + L_2 \cos(\theta_1 + \theta_2) + L_3 \cos(\theta_1 + \theta_2 + \theta_3), \\
y &= L_1 \sin\theta_1 + L_2 \sin(\theta_1 + \theta_2) + L_3 \sin(\theta_1 + \theta_2 + \theta_3), \\
\phi &= \theta_1 + \theta_2 + \theta_3.
\end{aligned}
$$

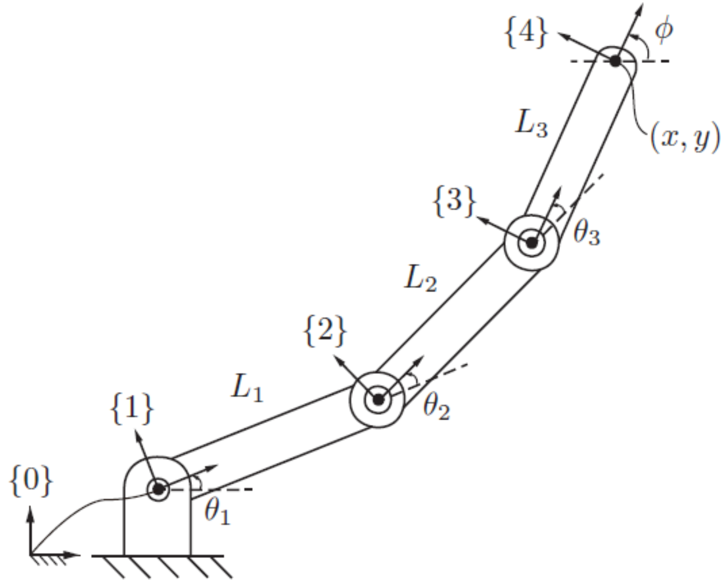


- Forward Kinematics
  - $T_{04} = T_{01}T_{12}T_{23}T_{34}$

$$
T_{01} = \begin{bmatrix} \cos\theta_1 & -\sin\theta_1 & 0 & 0 \\ \sin\theta_1 & \cos\theta_1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad
T_{12} = \begin{bmatrix} \cos\theta_2 & -\sin\theta_2 & 0 & L_1 \\ \sin\theta_2 & \cos\theta_2 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}
$$

$$
T_{23} = \begin{bmatrix} \cos\theta_3 & -\sin\theta_3 & 0 & L_2 \\ \sin\theta_3 & \cos\theta_3 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad
T_{34} = \begin{bmatrix} 1 & 0 & 0 & L_3 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}.
$$

FK: Matrix Exponential Representation (1/3)



- Zero(Home) Pose: M

$$M = \begin{bmatrix} 1 & 0 & 0 & L_1 + L_2 + L_3 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

- Consider each joint axis to be a zero-pitch screw axis
  - If θ_1, θ_2 are fixed at zero position, then the screw axis of θ_3 rotation in {0} frame
  - A turntable rotating with an angular velocity of $\omega_3 = 1\frac{rad}{s}$ about the axis of joint 3:
  - The linear velocity v_3 of the point on the turntable at the origin of $\{0\}$ is in the $-\hat{y}_0$ direction at a rate of $L_1 + L_2$ units/s.
  - $v_3 = -\omega_3 \times q_3$, $q_3 = [L_1 + L_2, 0, 0]$

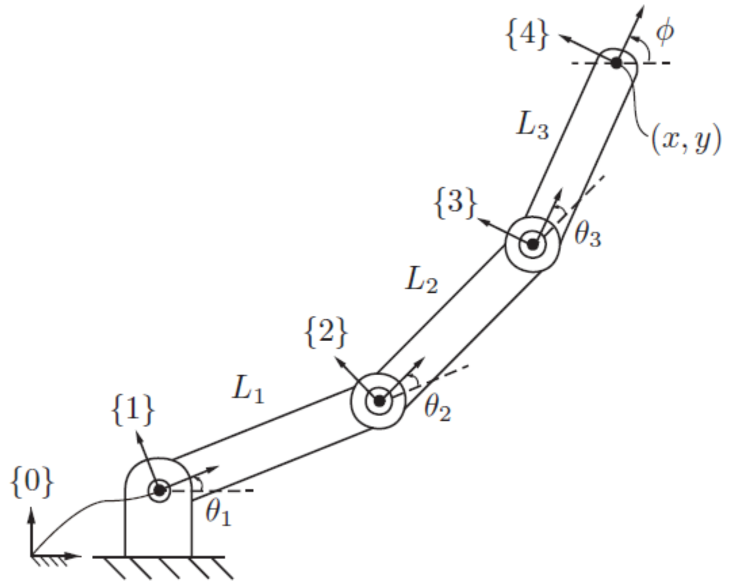
$$\mathcal{S}_3 = \begin{bmatrix} \omega_3 \\ v_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \\ -(L_1 + L_2) \\ 0 \end{bmatrix}$$

FK: Matrix Exponential Representation (2/3)

- The screw axis of S_3

$$\mathcal{S}_3 = \begin{bmatrix} \omega_3 \\ v_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \\ -(L_1 + L_2) \\ 0 \end{bmatrix} \qquad [\mathcal{S}_3] = \begin{bmatrix} [\omega] & v \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & -(L_1 + L_2) \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

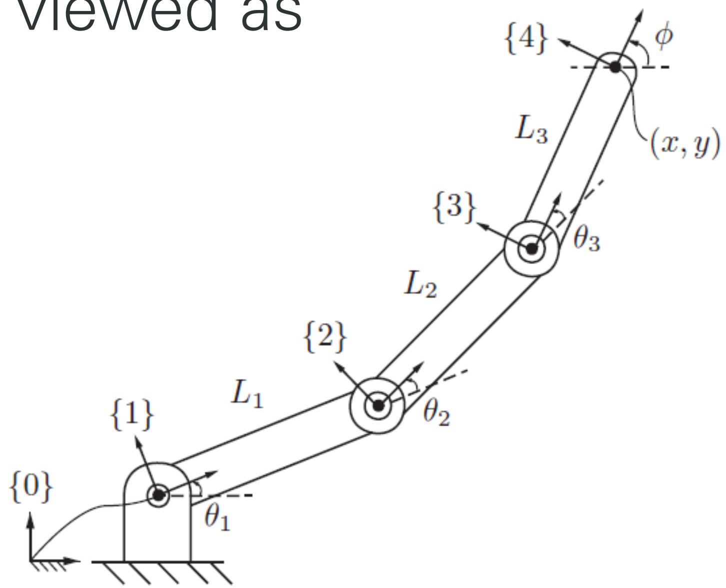
- Matrix exponential representation for screw motions
  - $T_{04} = e^{[S_3]\theta_3} M$ (for $\theta_1 = \theta_2 = 0$)



FK: Matrix Exponential Representation (3/3)

- For $\theta_1 = 0$ and any fixed θ_3, rotation about joint 2 can be viewed as applying a screw motion to the rigid (link2)/(link3) pair
  - $T_{04} = e^{[S_2]\theta_2} e^{[S_3]\theta_3} M$ (for $\theta_1 = 0$)

$$[\mathcal{S}_2] = \begin{bmatrix} 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & -L_1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$



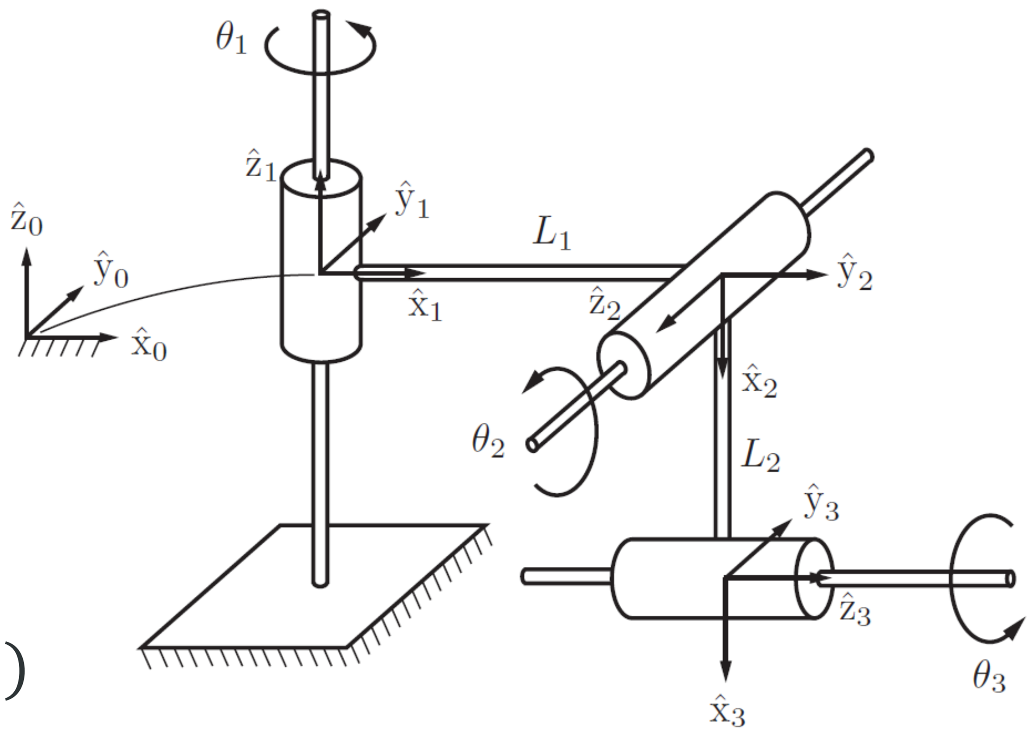
- Finally, keeping θ_2 and θ_3 fixed, rotation about joint 1 can be viewed as applying a screw motion to the entire rigid three-link assembly.
  - $T_{04} = e^{[S_1]\theta_1} e^{[S_2]\theta_2} e^{[S_3]\theta_3} M$ (for $\theta_1 = 0$) : This is called space form of PoE

$$[\mathcal{S}_1] = \begin{bmatrix} 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Example: Screw axes in base frame

- Forward kinematics
  - $T(\theta) = e^{[S_1]\theta_1} e^{[S_2]\theta_2} e^{[S_3]\theta_3} M$
- Zero position M
  - End-effector frame configuration in {0}

$$M = \begin{bmatrix} 0 & 0 & 1 & L_1 \\ 0 & 1 & 0 & 0 \\ -1 & 0 & 0 & -L_2 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$



- $S_1 = (\omega_1, v_1)$ where $\omega_1 = (0,0,1)$ and $v_1 = (0,0,0)$
- $S_2 = (\omega_2, v_2)$
  - $\omega_2 = (0, -1, 0)$
  - $v_2 = -\omega_2 \times q_2 = (0,\ 0,\ -L_1)$ where $q_2 = (L_1,\ 0,\ 0)$
- $S_3 = (\omega_3, v_3)$
  - $\omega_3 = (1,\ 0,\ 0)$
  - $v_3 = -\omega_3 \times q_3 = (0,\ -L_2,\ 0)$ where $q_3 = (0,\ 0,\ -L_2)$

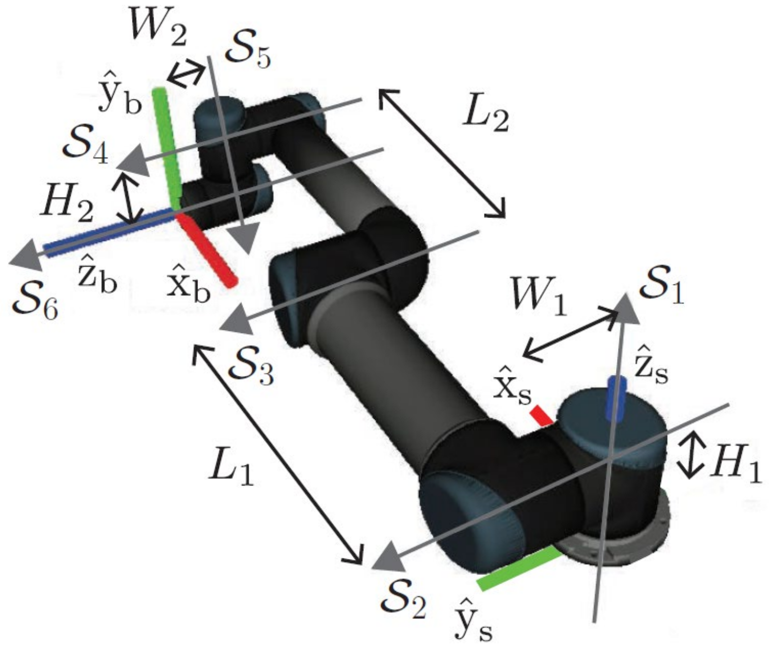
Example: Screw axes in the end-effector frame

- Note that
  - $e^{M^{-1}PM} = M^{-1}e^{P}M$ or equivalently $e^{P}M = Me^{M^{-1}PM}$
  - You should be able to prove this
  - $A = M^{-1}PM,\ e^{A} = I + A + \frac{A^2}{2!} + \frac{A^3}{3!} + \cdots = e^{M^{-1}PM} = I + M^{-1}PM + M^{-1}\frac{P^2}{2!}M + M^{-1}\frac{P^3}{3!}M + \cdots = M^{-1}e^{P}M$
- Previously, Space Form of PoE $T(\theta) = e^{[S_1]\theta_1} \dots e^{[S_{n-1}]\theta_{n-1}}\, e^{[S_n]\theta_n}M$
  - The end–effector configuration $M \in SE(3)$ when the robot is at zero/home position
  - The screw axes $S_1, \cdots, S_n$ expressed in the fixed base frame, corresponding to the joint motions when the robot is at its home position
- Using $Me^{M^{-1}PM} = e^{P}M$,
  - $T(\theta) = e^{[S_1]\theta_1} \dots e^{[S_{n-1}]\theta_{n-1}}\, \boxed{e^{[S_n]\theta_n}M} = e^{[S_1]\theta_1} \dots e^{[S_{n-1}]\theta_{n-1}}\, \boxed{Me^{M^{-1}[S_n]M\theta_n}}$
  - Which will lead to
- Body form of PoE
  - $T(\theta) = Me^{M^{-1}[S_1]M\theta_1} \dots e^{M^{-1}[S_{n-1}]M\theta_{n-1}}\, e^{M^{-1}[S_n]M\theta_n} = Me^{[B_1]\theta_1} \dots e^{[B_{n-1}]\theta_{n-1}}\, e^{[B_n]\theta_n}$
  - $[B_i] = M^{-1}[S_i]M$ the screw axes in the end–effector (body) frame when the robot is at the zero pose.

UR5 Example: Space From of PoE (1/2)

- UR5 dimension
  - $W_1 = 109mm, W_2 = 82mm, L_1 = 425mm, L_2 = 392mm, H_1 = 89mm, H_2 = 95mm$
- Zero position
  - Notice that $\hat{x}_b = -\hat{x}_s, \hat{y}_b = \hat{z}_s, \hat{z}_b = \hat{y}_s$
  - $M = \begin{bmatrix} -1 & 0 & 0 & L_1 + L_2 \\ 0 & 0 & 1 & W_1 + W_2 \\ 0 & 1 & 0 & H_1 - H_2 \\ 0 & 0 & 0 & 1 \end{bmatrix}$
- Screw axes $S_i = (\omega_i, v_i)$

i	ω_i	v_i
1	$(0,0,1)$	$(0,0,0)$
2	$(0,1,0)$	$(-H_1,0,0)$
3	$(0,1,0)$	$(-H_1,0,L_1)$
4	$(0,1,0)$	$(-H_1,0,L_1+L_2)$
5	$(0,0,-1)$	$(-W_1,L_1+L_2,0)$
6	$(0,1,0)$	$(H_2-H_1,0,L_1+L_2)$



UR5 Example: Space From of PoE (2/2)

- UR5 dimension
  - $W_1 = 109mm, W_2 = 82mm, L_1 = 425mm, L_2 = 392mm, H_1 = 89mm, H_2 = 95mm$
- What is the FK ? when
  - $\theta_2 = -\frac{pi}{2}, \theta_5 = \frac{\pi}{2}$
  - All others are zero

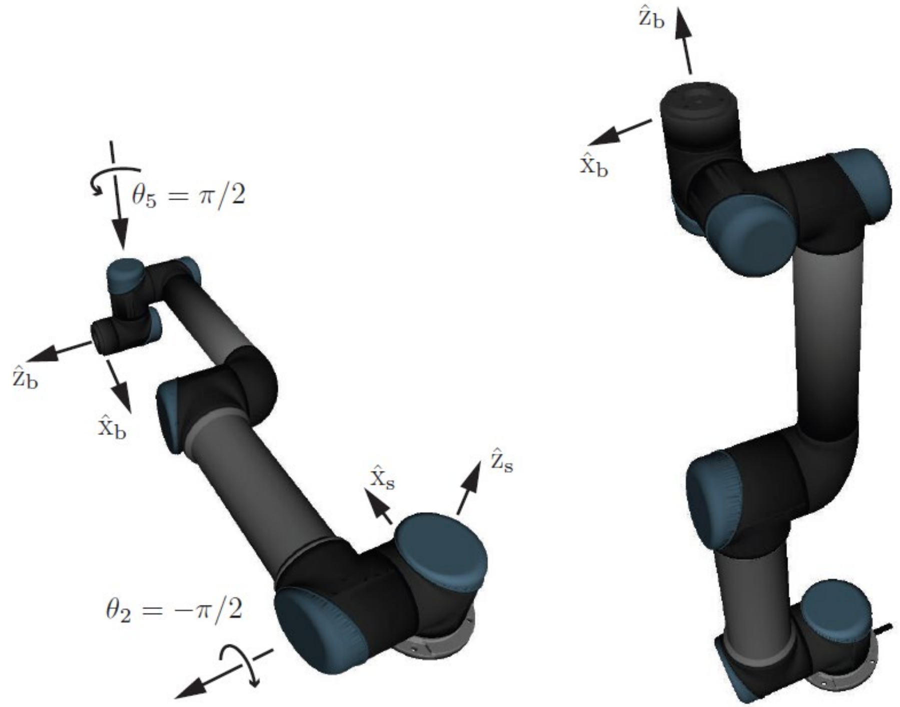
i	ω_i	v_i
1	$(0,0,1)$	$(0,0,0)$
2	$(0,1,0)$	$(-H_1,0,0)$
3	$(0,1,0)$	$(-H_1,0,L_1)$
4	$(0,1,0)$	$(-H_1,0,L_1+L_2)$
5	$(0,0,-1)$	$(-W_1,L_1+L_2,0)$
6	$(0,1,0)$	$(H_2-H_1,0,L_1+L_2)$

$$\begin{aligned} T(\theta) &= e^{[\mathcal{S}_1]\theta_1} e^{[\mathcal{S}_2]\theta_2} e^{[\mathcal{S}_3]\theta_3} e^{[\mathcal{S}_4]\theta_4} e^{[\mathcal{S}_5]\theta_5} e^{[\mathcal{S}_6]\theta_6} M \\ &= I e^{-[\mathcal{S}_2]\pi/2} I^2 e^{[\mathcal{S}_5]\pi/2} I M \\ &= e^{-[\mathcal{S}_2]\pi/2} e^{[\mathcal{S}_5]\pi/2} M \end{aligned}$$

$$e^{-[\mathcal{S}_2]\pi/2} = \begin{bmatrix} 0 & 0 & -1 & 0.089 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0.089 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$e^{[\mathcal{S}_5]\pi/2} = \begin{bmatrix} 0 & 1 & 0 & 0.708 \\ -1 & 0 & 0 & 0.926 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

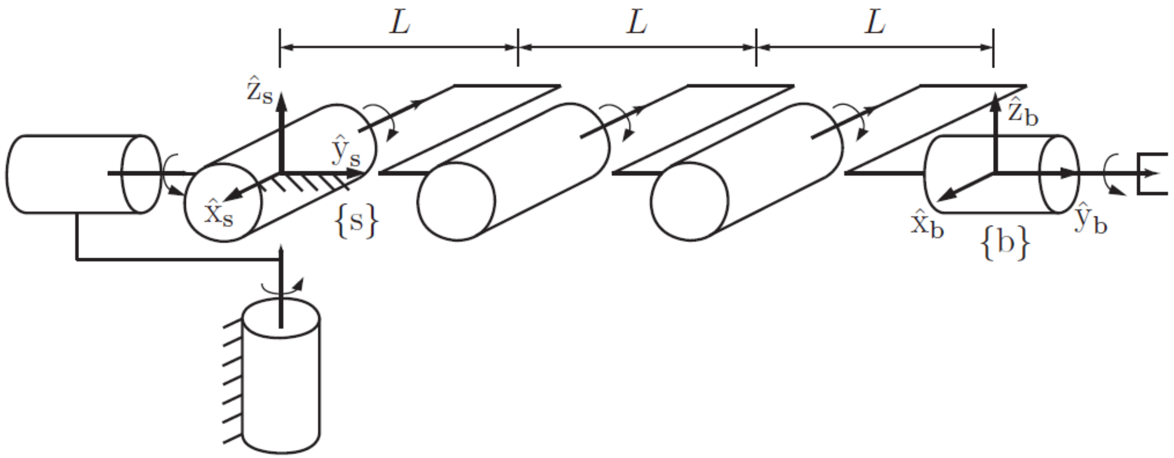
$$T(\theta) = \begin{bmatrix} 0 & -1 & 0 & 0.095 \\ 1 & 0 & 0 & 0.109 \\ 0 & 0 & 1 & 0.988 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$



Example: 6R

- Zero position

$$M = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 3L \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$



- Space form

i	ω_i	v_i
1	$(0,0,1)$	$(0,0,0)$
2	$(0,1,0)$	$(0,0,0)$
3	$(-1,0,0)$	$(0,0,0)$
4	$(-1,0,0)$	$(0,0,L)$
5	$(-1,0,0)$	$(0,0,2L)$
6	$(0,1,0)$	$(0,0,0)$

- Body form

i	ω_i	v_i
1	$(0,0,1)$	$(-3L,0,0)$
2	$(0,1,0)$	$(0,0,0)$
3	$(-1,0,0)$	$(0,0,-3L)$
4	$(-1,0,0)$	$(0,0,-2L)$
5	$(-1,0,0)$	$(0,0,-L)$
6	$(0,1,0)$	$(0,0,0)$

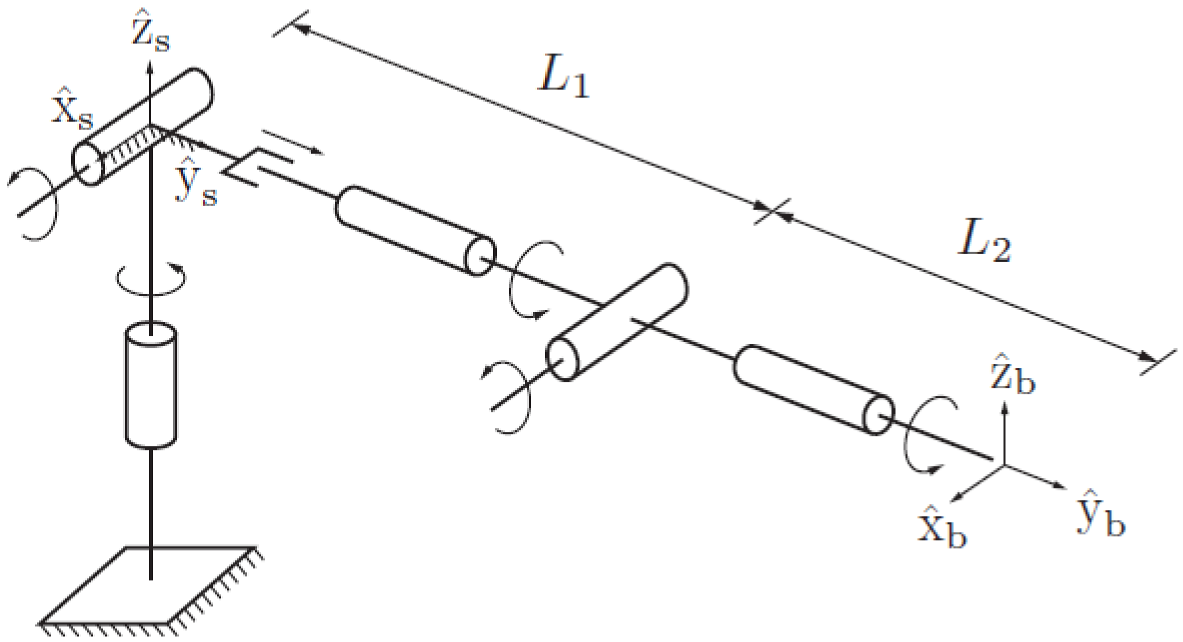
Example RRPRRR: Space form PoE

- Zero position

$$M = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & L_1 + L_2 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

- Screw axes $S_i = (\omega_i, v_i)$

i	ω_i	v_i
1	$(0,0,1)$	$(0,0,0)$
2	$(1,0,0)$	$(0,0,0)$
3	$(0,0,0)$	$(0,1,0)$
4	$(0,1,0)$	$(0,0,0)$
5	$(1,0,0)$	$(0,0,-L_1)$
6	$(0,1,0)$	$(0,0,0)$



Joint 3 is a prismatic